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RC4 → key generation

* RC4 is a Symmetric cipher, So both sides already share the same secret key

So The key generation in RC4 usually refers to: key scheduling Algorithms [KSA]

$S = [0,1,2,3,4,5,6,7]$ key $= [1,2,3]$

iteration 1
~~first step~~ $i = 0$, $j = 0$

$J = (j + S[i] + key[i \bmod keylength]) \bmod 8$

$\therefore$ ~~$j = 0$~~ $\therefore J = 1$

Swap $S[0] \leftrightarrows S[1]$

New array → $[1,0,2,3,4,5,6,7]$

iteration 2 $i = 1$ $j = 1$

$j = (1 + S[1] + 2) \bmod 8 = 3$

$\therefore S[1] \leftrightarrows S[3]$

new array → $[1,3,2,0,4,5,6,7]$

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iteration 3

swap S[2] → S[0]
new array [2, 3, 1, 0, 4, 5, 6, 7]

iteration 4
swap S[3] → S[1]
new array [2, 0, 1, 3, 4, 5, 6, 7]

Iteration 5
swap S[4] → S[7]
new array [2, 0, 1, 3, 7, 5, 6, 4]

iteration 6
swap S[5] → S[7]
new array [2, 0, 1, 3, 7, 4, 6, 5]

iteration 7
swap S[6] → S[6]
new array [2, 0, 1, 3, 7, 4, 6, 5]

Iteration 8
swap S[7] → S[5]
new array [2, 0, 1, 3, 7, 5, 6, 4]

③

PRGA formula

$i = (i+1) \bmod 8$ $\quad j = (j + S[i]) \bmod 8$

first Round

$S[2, 0, 1, 3, 7, 5, 6, 4]$

$i = 0$ $\quad j = 0$

$i = (0+1) \bmod 8 = 1$

$j = (0+0) \bmod 8 = 0$

Swap $S[1] \rightarrow S[0] \rightarrow [0, 2, 1, 3, 7, 5, 6, 4]$

$K = S[(S[1] + S[0]) \bmod 8]$

$k = S[(2+0) \bmod 8] = S[2] = 1$

Encryption

Message = 5

$\therefore C = M \text{ XOR } k$

$C = 5 \text{ XOR } 1 = 4$

Ciphertext = 4

El Gamal

prime — generator — private key

$p = 23$ $\quad g = 5$ $\quad x = 6$

Publick key

$$y = g^x \bmod p = 8$$

$\therefore$ Public key $(p, g, y) = (23, 5, 8)$

private key $\Rightarrow x = 6$

- Encryption

$m = 10$ $\qquad k = 3$

$c_1 = g^k \bmod p \longrightarrow 10$

$c_2 = m \times y^k \bmod p \longrightarrow 14$

$\therefore$ Ciphertext $(c_1, c_2) = (10, 14)$

- Decryption

$m = c_2 \times (c_1^{\,x})^{-1} \bmod p \longrightarrow 10$

## [1] ECDH

Alice side

Choose prv key:

$d_A = 5$

pub Key

$Q_A = d_A \times G = 5G$

Bob side

prv Key

$d_B = 9$

Pub key

$Q_B = d_B \times G = 9G$

Shared Secret Computation

Alice Computes

$S = d_A \times Q_B$

$S = 5 \times 9G$

$S = 45G$

Bob Computes

$S = d_B \times Q_A$

$S = 9 \times (5G)$

$S = 45G$

ALADIB

## [2] KDF

input : Shared secret x-coordinate

$x = 45$

Salt = 11

info = "Session"

### [1] extract

$pRK = HMAC - MD5(Salt, x)$

$pRK = HMAC - MD5(11, 45)$

output → $pRK = A1B2C3D4$

### [2] expand

$OKM = HMAC - MD5(pRK, info \| 01)$

$OKM = HMAC - MD5(A1B2C3D4, \text{"Session01"})$

output → $OKM = \underline{9F86D081884C7D65}$

Final derived Key

الموضوع:

التاريخ: / /

Twofish

Key generation

1. master key 128 bit

K = A1B2C3D4E5F60708112233
44 55 667788

2- split into 32-bit words

$M_0$ = A1B2C3D4
$M_1$ = E5F60708
$M_2$ = 11223344
$M_3$ = 55667788

3- Separate Even/odd words

$Me$ = [$M_0$, $M_2$] $Mo$ = [$M_1$, $M_3$]

$Me$ = [A1B2C3D4, 11223344]
$Mo$ = [E5F60708, 55667788]

4- Generate Round Key

$A = h(0, Me)$
$A = 10203040$

$B = h(1, Mo)$
$B = FFEEDDCC$

5- First Round subkey

$K_0 = A + B$
$K_0 = 0F1E0E0C$
$K_{2i+1} = ROTL(A + B \times 2, 9)$
$K_1 = 4A5B6C7D$

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Encryption

$P = 123456789ABCDEF0$

$L_0 = 12345678$
$R_0 = 9ABCDEF0$

2. F function

$$F(R_0, K_1) = R_0 \oplus K_1$$

$$9ABCDEF0 \oplus 4A5B6C7D$$

$$= D0E7B28D$$

Rotate left $= A1CF651B$

3. feistel update

$$L_1 = R_0$$

$R_1 = L_0 \oplus F(R_0, K_1)$
$= 12345678 \oplus A1CF651B$
$= B3FB3363$

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$L_1 = 9ABCDFF0$

$R_1 = B3FB3363$

$C = 9ABCDFF0B3FB3363$

الموضوع:

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MD5

1- intial Registers

A= 67452301
B= EFCDAB89
C= 98BADCFE
D= 10325476

2- message block

abc

after padding $M_0 = 61626380$

3- MD5 function

$f(B,C,D) = (B \wedge C) \vee (\neg B \wedge D)$

F= 98BADCFE

4- Add constants

Constant → $K_0$ = D76AA478
shift amount → S = 7

$$TEMP = A + f + M_0 + K_0$$
$$= 57CE06D8$$

5- Left Rotation

$$ROTL(TEMP, S) \quad 7$$
$$= E7036C2B$$

6- Addition

$$A = B + E7036C2B$$
$$= D6D117B4$$